

The One Introduction Problem: Round 1

Research Note

Team	solo
Version date	8 October 2026
Repository	https://github.com/Sga-HQ/The-Sequential-Matching-Problem-Hackathon_round1
Code and results behind every number	commit d86ff40ecc561157a6c9b15f6b977acc0e556752
Organiser kit	participant specification 1.0.0, kit commit a8e26b35118cfa8e886a02f93984923e43ab64f6

How to read the claims. Each claim is labelled by the kind of evidence behind it:

- **[rule]**: the organisers' specification.
- **[code]**: behaviour we read in the public simulator source (`kit.py` at the kit commit above).
- **[dev]**: an empirical result on our development seeds.
- **[oracle]**: an offline diagnostic that uses hidden simulator values (never available to the policy).
- **[motivation]**: a literature result that motivates a choice without proving it applies here.
- **[plan]**: a Round 2 experiment.

All uncertainties written $\pm$ are **one paired standard error across episodes**. 95% intervals are about ± 2 SE.

0. Summary

We treat the challenge as **sequential constrained allocation** rather than pair-by-pair compatibility prediction. Each day the policy must:

- decide which unknown fields are worth clarifying;
- then build a valid, non-overlapping matching using only what it can observe.

What we measured.

- **Reciprocal feasibility and information availability dominate.** Removing clarification reduced realised MSMI by about **72%** [dev, 240 episodes].
- **Reciprocally eligible alternatives are few** (about 3 per person over an episode; definition in §1.4). The kit's greedy baseline already introduces about **75%** of those pairs [oracle].
- So better ranking has limited opportunity unless clarification first exposes the relevant information. Much of the remaining pair-level variation comes from latent stochastic terms that the policy cannot see [code, oracle].

Status: prototype vs evidence-supported candidate.

	Tested prototype (first build)	Evidence-supported Round 2 candidate
Feasibility	strict two-way filtering, batch validation	same
Hard clarification	dealbreaker bundles, kit order	same
Soft clarification	1-point questions, broad	broad (anyone with ≥ 1 feasible partner); the margin rule was tested and rejected (§9.6)
Scorer	online Beta learner, Thompson draws, personal reply and yes terms	fixed offline scorecard (posterior means, no online updates, no personal terms)
Matching	most pairs first, then best total, $\times$ coverage boost 1.5 $\times$ degree boost	most pairs first, then best total; no coverage boost ; degree boost provisional (no measured effect)
Feedback	updates the learner	kept per stage for bookkeeping and availability; not used to retrain (it changed decisions without improving them, §9.6)

Development results (hidden-information expected-wins diagnostic, §1.4: not the official metric, but shown there to have the same expectation) [dev].

- **Prototype vs greedy on three disjoint blocks:** +2.7%, +6.3% and +0.2%. Episode-weighted paired mean over 180 episodes: $+0.0113 \pm 0.0028$ per 100 (95% CI $+0.0057$ to $+0.0168$). Random-effects estimate: $+0.0104 \pm 0.0048$. The blocks clearly differ ($Q = 7.8$, 2 degrees of freedom).
- **Candidate vs greedy on a fourth, new block** (120 episodes): **+0.0205 per 100 (+5.5%)**, ± 0.0051 cluster-robust standard error, about 4 SE.
- **The one scenario family with a gain in every block is cold start** ($+0.023 \pm 0.006$ over the first three blocks), where information is scarcest.
- **Realised MSMI does not yet confirm any gain.**
 - Signs vary across blocks.
 - On the fourth block the greedy baseline scored higher on realised MSMI (-0.096 ± 0.050 , cluster-robust). On that block greedy's realised MSMI was 0.46 against its expected 0.37, while ours was close to its expectation.
 - **Checked (8 Oct):** the expected-wins formula reproduces the simulator's own outcome code exactly (360,000 replayed introductions, simulated/formula = 1.003). Across all 1,980 development episodes the realised-minus-expected gap is -0.037 ± 0.039 per 100 (cluster-robust over the 50 seeds, $z = -1.0$), which is consistent with chance. So the block-7400 disagreement is outcome noise, not a diagnostic bias (§1.4).
 - With 120 episodes, realised MSMI can only detect gains of roughly 38% or more (80% power), while expected wins can detect about 4%. The confirmation run (§9.7) therefore reports both, and is sized for realised MSMI as well.

What remains open.

- A relaxed offline oracle suggests limited remaining opportunity from ranking alone. We treat it as an optimistic, simulator-specific diagnostic, **not a formal bound** on all sequential policies.

- **Open question 1: dynamic option value.** What does occupying a person for 8+ days cost others in a thin, changing graph? A one-line opportunity-cost penalty added nothing measurable beyond removing the coverage boost (\$9.4). Short look-ahead (one-day rollout) and selective waiting are untested.
- **Open question 2: decision-relevant use of feedback.** Online learning from replies changed 9–15% of daily matchings but did not improve expected wins. Shift-specific adaptation and asking rules that value answers by the decisions they change are the remaining candidates (H7).
- Round 2 will freeze the policy, then evaluate it once on a pre-declared, participant-controlled seed block (\$9.7), reporting realised MSMI, coverage, mutual acceptances, clarification cost, waiting time, runtime and invalid-episode counts.

1. Problem interpretation

1.1 Daily decision [rule]

Each day the policy makes two decisions.

Ask. Spend up to 12 points: the dealbreaker bundle costs 3 (all hard fields at once); a named soft field costs 1. Answers arrive immediately.

Introduce. Return a batch of pairs. Every pair must satisfy:

- all reciprocal hard constraints;
- availability of both people;
- no repeated pair;
- no person used twice in the batch.

The whole batch is validated atomically. An empty batch is valid. One invalid assessed episode makes the submission ineligible for technical ranking, so **validity comes first**.

1.2 One introduction's journey [code: `Simulator.advance`]

```
Day t      Introduced; both occupied.
Day t+1..t+7 Each person replies yes/no on a random day, or not at all (a per-person
reply probability, 0.55–0.98).
  └ any "no" or no reply → ends; the pair can never be repeated; both available again
from day t+8
  └ both "yes" → date 1–14 days after the later reply (+5–12 days in the delayed
scenario);
    both occupied until 6 days after the date
    └ the date does not happen (22%) → ends
    └ the date happens → each answers "second date?" 1–5 days later, or not at all
      (same per-person reply probability as above)
      └ both "yes" → both paused. Counts for MSMI only if both answers arrive
within 3 days
          and the date is ≤ 30 days after the introduction [rule].
```

The 8-day occupancy is set at introduction for both people, whatever their answer.

1.3 The pool (development scenario, 10 worlds) [dev]

Group	People per world	Can be introduced?
All hard fields known on arrival	69.9	yes
Hard fields unknown but askable (each misses 7-11 of 11; none misses only 1-2)	65.4	after one 3-point ask
Declined at least one hard field	64.6	never (declined answers are never inferred) [rule]

About 70% arrive on day 0 and the rest by day 20. About 12% leave on a day drawn uniformly from 22-60, fixed when the person is generated and independent of their profile [code: generate].

1.4 Binding constraints

Definition (used in §0 and here). An **allowed pair** is a unique unordered pair that satisfies:

- both people arrived by day 59;
- neither declined a hard field;
- the pair satisfies all reciprocal hard constraints **after full revelation of hidden hard answers**;
- *regardless* of whether both people are ever simultaneously available.

This is an offline diagnostic [oracle], computed for the development scenario on 10 worlds (experiments/theory_ceilings.py).

Constraint	Measurement	Consequence
Unknown hard fields	no-asks baseline: realised MSMI –72% (240 episodes) [dev]	Clarification is essential
Few allowed pairs	124 per world, ≈ 3 per person; greedy introduces 92 of them (75%; 98% in sparse) [oracle]	Ranking matters only where a choice exists
Choice exists mainly early	after day 20, ~ 105 people are free and fully known each day, but only $\sim 1-2$ currently feasible pairs exist among them [dev]	Ranking matters on day 0 and while newcomers arrive
Why allowed pairs go unused	53% one person already paused · 27% one person left · 20% one person was occupied [oracle]	Only the last two are addressable (timing, option value)
Outcome noise	~ 1 qualifying outcome per episode; detecting $1.0 \rightarrow 1.5$ needs ~ 79 episodes per policy [dev]	Compare with a low-variance diagnostic (below), then confirm on MSMI

Expected wins (development diagnostic, not the official metric).

- For each introduced pair, we compute its success probability from the simulator's outcome formula, using hidden per-person values and integrating over the shared pair term with 5-point Gauss-Hermite quadrature. We then sum these probabilities.
- **Numerical check:** the 5-point integration was compared with a 60-point Gauss-Hermite reference on 3,000 random pairs from development worlds. Maximum absolute difference 4.1×10^{-7} ; mean relative difference 0.001% [dev,

checks_v3.py]. This validates the numerical integration; the completeness of the formula is checked next.

- **Delayed scenario: 30-day date rule.** The date delay is $\max(\text{two reply delays } U\{1..7\}) + U\{1..14\} + U\{5..12\}$, drawn independently of the pair, the day and the policy [code]. So $P(\text{date within 30 days})$ is an exact constant: $5,368 / 5,488 = 0.9781$. The diagnostic now multiplies by it (fourth block onwards). Earlier blocks overstate delayed-scenario expected wins, and their differences, by exactly 2.2%. Relative comparisons are unaffected.
- **Check against outcomes:** over 840 fourth-block episodes, mean expected wins 0.386 vs realised MSMI 0.357 per 100. Per-policy realised minus expected values vary in sign across blocks for every policy, including greedy, so we see no sign of a policy-specific bias in the diagnostic.
- **Exactness check against the simulator** [dev, experiments/pwin_check.py]. We replayed the kit's own `advance()` and `metrics()` code on 1,200 random pairs (eligibility does not enter the outcome code, so it was bypassed; 200 per scenario family, assignment days 0, 10 and 40), each under 300 independent outcome seeds: 360,000 introductions. Simulated wins 3,756 vs formula 3,743.4 (ratio 1.003). Per family, the ratios were 0.97–1.03 and every $|z| < 0.9$. Combined with the code reading (outcomes are drawn once at assignment from a stream keyed on seed, pair and day; leaving after an introduction does not affect its outcome; every outcome lands before day 100), this means the per-introduction formula is the simulator's exact success probability.
- **Why that makes it an unbiased estimate of MSMI** [code + algebra]. Write realised MSMI as a sum over introductions of win indicators W_i . Each introduction's luck is a fresh random stream that the policy cannot have seen when it chose the pair (no pair can be repeated). So $E[W_i \mid \text{everything up to assignment}] = p_i$, the formula's value, even though the policy adapts to earlier feedback. By the tower property, $E[\sum p_i] = E[\sum W_i]$: summed expected wins has the same expectation as realised MSMI for any policy, adaptive or not. It removes only the outcome-luck variance, which is why its paired standard errors are about $10\times$ smaller. Across all 1,980 development episodes the realised-minus-expected gap is -0.037 ± 0.039 per 100 (cluster-robust over the 50 seeds, $z = -1.0$), consistent with this.
- It is unavailable to the policy. We use it only to reduce outcome noise when selecting candidates. Claims of improvement are judged on untouched seeds (§9.7).

2. Hypotheses

- **H1 (clarification).** Spending otherwise-unused budget on soft-field questions reduces blank clues and improves scorer discrimination. It improves wins only where a choice exists.
- **H2 (reply behaviour).** Because each person's reply probability applies at both the introduction and the second-date stage [code], down-weighting people with a record of non-reply should improve expected wins. *Not supported so far (§9.5).*
- **H3 (global allocation).** Lexicographic maximum-cardinality, maximum-weight matching avoids known same-day failures of best-pair-first selection. Whether this produces measurable episode-level gains in the thin public simulator is an empirical question.

- **H4 (limited discounting).** For stationary public scenarios, equal weighting minimises variance. The drift scenario shifts the common intercept on day 35, so equal weighting can bias absolute probabilities. $\gamma = 1$ is a default, not a proved optimum (§5.5). It is moot for the candidate, which does not learn online.
 - **H5 (headroom).** Under our relaxed oracle, sequential scarcity leaves roughly $\leq 40\%$ improvement over the public baseline; oracle soft-field ranking alone leaves about 8%. We will challenge these values with stronger oracle policies, and check whether omitted timing or assignment mechanisms break the relaxation.
 - **H6 (option value, open).** Accounting for what an introduction takes from other people's future options (8+ day occupancy, departures, arrivals) beats same-day matching.
 - **H7 (closed-loop feedback, partly tested).** Post-introduction events help only if their stage-specific updates change later decisions for the better before the horizon ends. Test so far (§9.6): population field learning and personal learning changed 9% and 15% of daily matchings, with no improvement in expected wins. Remaining Round 2 tests: shift-specific adaptation, and asking rules valued by changes to the full matching.
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3. Reciprocal feasibility

Definition. For hard rule k , let $c_{k(i \rightarrow j)} \in \{0,1\}$. A pair is feasible on day t when

$e_{ij}(t) = \prod_k c_{k(i \rightarrow j)} \cdot c_{k(j \rightarrow i)} \cdot [\text{both available, arrived, fully known, same pool}] \cdot [\text{never introduced}]$.

- Unknown values never pass. Declined values are never inferred [rule].
- **Implementation:** each rule is evaluated as a 0/1 column over all people (numpy), and the columns are combined with logical AND.
- **At the evaluated scale (200 people)** the daily filter takes ≤ 26 ms, and the slowest full simulated day took 45 ms in-process [dev]. A scaling benchmark (Appendix A) shows the column method is the fastest of four equivalent implementations, at no cost at small scale.

Batch validation before output. Every pair:

- passes the kit's official `eligibility()`;
- has two distinct IDs that are arrived, available and in the same pool;
- has not been introduced before.

Also, no member appears twice in the batch, the ask budget is respected, and output is plain JSON (no NumPy types, NaN or Infinity). Local tests additionally pass every proposed batch through the public simulator's validator. All of our development episodes ($> 1,500$) ran with **0 invalid actions**.

4. Clarification policy

Measurements.

- The kit baseline spends 199 of 720 points per episode [dev]; after about day 17 no askable hard-field gaps remain.
- An oracle test of dealbreaker-ask *order* (180 episodes per asker) found differences $\leq \sim 1\%$ in both-said-yes, within noise [dev].
- About 62% of soft clues are blank at decision time (82% in cold start) [dev, 15 worlds $\times$ 6 scenarios].

Rules (a decision-relevance heuristic, not a full value-of-information computation):

1. **Hard bundle (3 points)**, in kit order, for available askable people. In Round 2 we will prioritise people whom ready members' stated preferences already accept, a cheap exact half-check using always-visible age, gender and zone.
2. **Soft fields (1 point each) with the remaining budget**, in field order goal $\rightarrow$ pace $\rightarrow$ lifestyle $\rightarrow$ conversations.
 - **Broad rule (current):** ask any available, fully known person with ≥ 1 currently feasible partner.
 - **Matching-margin rule (tested):** ask only if the person's second-best edge score is $\geq 0.6 \times$ their best. A soft answer moves odds by roughly $\times 0.7$ – $\times 1.4$, so it could then reverse their top choice.
3. **Never ask** declined, unavailable or paused people [rule].

We do not currently ask the three soft fields whose estimated effect was negligible in public development data (Information Value ≤ 0.009 in every tested scenario). We keep them in the observation pipeline so a shifted private scenario cannot cause a parsing or correctness failure.

Theory. Adaptive-submodularity results (Golovin & Krause, JAIR 2011) motivate greedy one-question-at-a-time querying when diminishing-return assumptions hold [motivation]. We do not claim our full matching objective satisfies those conditions. The simulator ablations are the evidence.

Rejected: filling blank answers with guesses from "similar users".

- Not allowed: "never fabricate a default preference"; declined answers "must not be inferred" [rule].
- Uninformative: soft fields are drawn independently per person [code].

5. Probability model (the scorer)

5.1 Target and operational score

By the chain rule, a qualifying outcome needs:

$P(\text{win}) = P(\text{both reply}) \cdot P(\text{both yes} \mid \text{replied}) \cdot P(\text{date}) \cdot P(\text{both answer again}) \cdot P(\text{both yes again}) \cdot P(\text{both on time, date} \leq 30 \text{ days})$

- **[code]** Each person's reply probability r_i is a single hidden value. The simulator uses it both for the introduction reply and for the second-date answer (`Simulator.advance`). The probability of the date, of an on-time answer, and the 30-day rule do not depend on visible fields.

- Our **operational ranking score** is

$$s(A, B) = \hat{P}(A \text{ yes} \mid \text{clues}, A) \cdot \hat{P}(B \text{ yes} \mid \text{clues}, B) \cdot \hat{r}_A^2 \cdot \hat{r}_B^2$$

The square follows from the inspected mechanism. Each person's $\hat{r}$ is estimated from both kinds of answer event.

- This is a **ranking approximation** to the full journey probability. We omit factors that we cannot estimate reliably online or that are constant across pairs under the inspected simulator: the shared pair term, the visible-goal term at the second stage (§5.3), and the date-timing distribution. The approximation is judged against expected wins, not assumed correct.

5.2 Measured conditional probabilities [dev: kit baseline, development, 30 worlds, 5,468 reply chances; fields as known at decision time]

Quantity	Value (n)
P(reply)	0.76 (5,468)
P(reply replied last time) / P(reply silent last time)	0.77 (2,361) / 0.68 (798)
P(yes replied)	0.48 (4,132)
P(yes yes last time) / P(yes no last time)	0.51 (815) / 0.44 (1,009)
P(yes goal same) / P(yes goal different)	0.61 (758) / 0.37 (718)
P(yes goal and pace both same)	0.70 (257); additive log-odds prediction 0.70
P(date both yes)	0.76 (400)
P(second yes date): goal same / different / unknown	0.63 (118) / 0.64 (73) / 0.65 (292)
P(second answer within 3 days answered)	0.60 (483)

Approximate binomial standard errors are $\sqrt{p(1-p)/n}$: about ± 0.02 for $n \approx 700$, and ± 0.04 – 0.06 for the second-stage rows.

Information Value per field (yes vs no; 15 worlds per scenario). Scale (Siddiqi 2006): < 0.02 negligible, 0.02 – 0.1 weak.

Field	development	shift	cold start
goal	0.074	0.006	0.050
pace	0.002	0.066	0.004
lifestyle	0.005	0.006 (direction reversed)	0.012
conversations	0.005	0.017	0.006
other three soft fields	≤ 0.002	≤ 0.009	≤ 0.008

5.3 What this implies

1. **Additive log-odds.** The inspected simulator and our conditional measurements support an additive log-odds model for the tested fields. We found no interaction with adequate support, and with ~ 25 early replies per world and a median of 2 introductions per person, interaction models, per-user vectors and tree ensembles are not justified.

2. **Selection at the second stage.** The simulator contains a visible-goal term at the second stage [code], yet it is invisible in observed data. Our interpretation: pairs with different goals reach a date mainly when their shared pair term is favourable, and that term also raises second-stage answers. Modelling the whole chain over all introductions is the standard remedy for this kind of post-selection bias (ESMM; Ma et al., SIGIR 2018) [motivation].
3. **Habits persist.** Reply behaviour and yes-propensity carry over between a person's introductions, but there are ~2 introductions per person, so strong shrinkage is needed.
4. **The informative field depends on the scenario** (goal normally, pace under shift). The prototype learned the weights online from an offline prior. The ablations (§9.5–9.6) favour a fixed offline scorecard, and adaptive weights stay only if a shift-specific test shows outcome value.

5.4 Estimator and reproducibility (prototype; the candidate keeps only the offline prior means)

Field effects. For each field and state (same / different), a Beta belief about the yes-rate.

- The prior mean comes from an offline fit on public training seeds; its strength is $k_{\text{field}} = 40$ pseudo-replies. Tuning showed 12 was worse (§9.3).
- An unknown clue contributes 0, so a blank is never a mismatch.

Reply behaviour. $r_i \sim \text{Beta}(0.76 \cdot 10 + \text{answered}_i, 0.24 \cdot 10 + \text{unanswered}_i)$.

- *Answered* and *unanswered* are counted over introduction responses and second-date answers.
- An unanswered event is recorded when the simulator reports `missing_reason = no_response`, on day $t+7$ for introductions and $\text{date}+5$ for second answers.

Yes-propensity. Person i 's yes-rate among their *answered* introduction replies, shrunk toward the base rate with 10 pseudo-replies. Non-response never enters it.

Thompson sampling.

- At each decision, the policy draws **one** value per population effect and **one** per person, and shares them across all candidate edges that day. There are no independent per-edge draws.
- The generator is seeded deterministically from the day and phase ($7919 \cdot \text{day} + \text{phase}$).
- All beliefs are recomputed from the observed feedback history on every call, so no random state or counts need to be carried in memory, and repeated runs are reproducible.
- Exploration happens only among feasible edges; **a hard constraint is never relaxed.**

Censoring. A reply that has not yet arrived is pending, not "no". An unanswered message updates reply behaviour only.

Calibration. Mean predicted $P(\text{yes})$ was 0.465 against an observed 0.466 (tuning round 1, broad asking) [dev]. Reliability curves and the Brier score are planned. (The KS

statistic measures separation, not calibration.)

AUC definition. All AUC values in this note are **directional introduction-response AUC**:

- the target is whether an introduced person answered yes;
- among answered replies, for introductions made by the policy under test;
- scored with that policy's posterior-mean prediction at decision time.

Because each policy is evaluated on its own introductions, AUCs are comparable only within one run table.

5.5 Discounting old replies (argument under the inspected mechanism)

- [code] Field effects, personal reply probabilities and yes-propensities are fixed within an episode in every public scenario. The shift scenario changes the weights from day 0; the drift scenario subtracts the same 0.5 from everyone's log-odds from day 35, which leaves each person's ordering of options unchanged.
- For a constant rate p estimated by a weighted mean with $\sum w_i = 1$, the variance is $p(1-p) \cdot \sum w_i^2$, and $\sum w_i^2 \geq 1/n$ with equality only for equal weights (Cauchy-Schwarz). In scenarios with no within-episode change, discounting therefore adds variance without reducing bias.
- The drift scenario breaks strict stationarity. Its day-35 intercept shift leaves each person's ordering of options unchanged, but equal weighting biases absolute probabilities, and products of two directional probabilities need not keep their exact order. So $\gamma = 1$ is a development default, not a general result.
- Where the environment does drift (a real service), exponential smoothing with $\gamma = 1 - \alpha$, where $\alpha = (-q + \sqrt{q^2 + 4q})/2$, is optimal for a local-level model with signal-to-noise ratio q (Muth, JASA 1960) [motivation]. $q = 0$ gives $\gamma = 1$.
- **Caveat.** If private scenarios contained within-episode drift in field effects, this choice would be wrong. An empirical check ($\gamma \in \{0.95, 0.98, 1\}$) is planned.

5.6 Offline ceilings for discrimination [oracle, 15 worlds per scenario]

Predictor	Directional AUC
Our policy's scorer (dev runs, by asking rule; §9.3–9.6)	0.54–0.60
True soft-field effects, all four fields known	0.61–0.65
+ each person's hidden yes-propensity	0.70–0.72
+ the shared pair term	0.74–0.75

Even with the inspected hidden traits, substantial variation remains stochastic. The gap between our scorer and the true-soft-field row comes mostly from blank clues, which is why clarification and scoring are designed together.

6. Allocation

Objective (lexicographic). Each day, first maximise $\sum x_{ij}$ (the number of introductions). Then, among maximum-cardinality matchings, maximise $\sum v_{ij} x_{ij}$, subject to $\sum_j x_{ij} \leq 1$ and $x_{ij} \in \{0,1\}$.

- Solved with networkx `max_weight_matching(maxcardinality=True)` (Edmonds' blossom algorithm).
- Weights are integers, $\lfloor 10^9 \cdot v \rfloor + 1$, so differences of about 10^{-9} in score are preserved. Ties are broken by networkx's deterministic processing of a deterministically built graph.
- **Why cardinality first:** every feasible introduction has positive expected value in the simulator, and our measurements show ranking differences are small relative to one more introduction.

Weights. $v_{ij} = s(i, j) \cdot u_i \cdot u_j$, where

- $u_i = c_{\text{new}}$ (if person i has never been introduced) $\times (1 + c_{\text{deg}} / d_i)$;
- d_i = number of currently feasible partners;
- the tested prototype used $c_{\text{new}} = 1.5$ and $c_{\text{deg}} = 0.5$. **The candidate uses $c_{\text{new}} = 1$** (the coverage boost removed). c_{deg} is provisional: removing it changed expected wins by $+0.0009 \pm 0.0018$ (§9.6).

Coverage is only a tie-breaker after MSMI and mutual acceptances [rule]. A multiplier that trades estimated primary value for coverage is therefore an objective mismatch. We removed it for that reason; the small favourable result (§9.4) is consistent with this but does not prove it on its own. Coverage may still break exact ties between matchings of practically equal value.

Why not best-pair-first. Kit example: best-pair-first gives 0.95; the best set gives 1.30 [rule]. Greedy can lose up to half the optimum.

Opportunity cost (H6, first test). $v'_{ij} = v_{ij} - \lambda \cdot (O_i^{(-j)} + O_j^{(-i)})$, where $O_i^{(-j)} = \sum_{k \neq j} v_{ik} / d_k$ is the value of person i 's other options, weighted by the scarcity of those partners. Result: §9.4.

Fairness. We considered and rejected upper score cutoffs. Real systems protect disadvantaged users with floors or quotas, not caps (FairRec, FA*IR). In dating apps, attention is spread because popular users become congested (Rios, Saban & Zheng, MSOM 2023) [motivation], and the simulator has no congestion. Details: Appendix C.

7. Missing and delayed data

Situation	Rule
Unknown hard field	Never passes; ask first
Declined hard field	Never introduced, never asked, never inferred
Unknown soft field	Contributes 0; ask if relevant
Reply not yet arrived	Pending, not "no" (right-censoring)
Reply never arrives	Updates reply behaviour only
Second-stage answers (up to ~26 days later)	Recorded per stage (below)
Member unavailable with no active introduction	Treated as currently non-actionable; we do not infer the reason
Delayed scenario	Longer occupancy; dates > 30 days after introduction cannot qualify (not controllable)

Each feedback event updates only the stage it belongs to.

Event	What it is evidence about	Prototype use	Candidate use
Introduction reply missing (day t+7)	that person's first-stage reply probability	reply habit	bookkeeping only
Introduction "no"	that person's acceptance of this partner (not reply failure)	field counts, yes-propensity	bookkeeping only
Introduction "yes"	reply happened; acceptance	field counts, yes-propensity, reply habit	bookkeeping only
Both yes, date did not happen	date occurrence (constant 0.78 for every pair [code])	–	occupancy timing
Second-stage answer missing	second-stage reply (same hidden reply probability [code])	reply habit	bookkeeping only
Second-stage "no" / "yes"	second-meeting preference	–	pause / occupancy
Late second-stage "yes" (> 3 days)	positive preference, but no MSMI credit	–	bookkeeping only

Population-level stage rates (date occurrence, on-time answering) are the same for every pair in the simulator [code]. Learning them online cannot change any ranking, so they were not tested as a learning arm. The arms that could change decisions (field effects, personal habits) were tested in §9.6.

Data per world (development, kit baseline): 172 reply opportunities (65 yes, 65 no, 42 none), 14 mutual acceptances, 11 dates, 2.3 pauses. Only ~25 replies arrive in the first 10 days, so the prior carries the early decisions [dev].

8. Existing work used, and why

Piece	Source	Status	Evidence here
Two-way hard filtering	Reciprocal recommenders (RECON; Pizzato et al., RecSys 2010)	implemented	required for validity
Column checks	column-store / bitmap index practice	implemented	Appendix A
Lexicographic max-cardinality, max-weight matching	Edmonds (1965); networkx 3.4.2	implemented	§6, §9
Additive log-odds scorer, WoE / IV	credit scorecards (Siddiqi 2006)	implemented	§5.2–5.3
Beta beliefs + Thompson sampling	Chapelle & Li (NeurIPS 2011); Russo et al. (2018); organiser session "Exploration, Exploitation and Online Learning" (6 Oct 2026)	tested; rejected (§9.5–9.6)	–
Whole-chain target	ESMM (Ma et al., SIGIR 2018)	motivation	§5.3
No discounting	Muth (1960)	argument under inspected mechanism	§5.5

Piece	Source	Status	Evidence here
Prompt matching in thin markets	Akbarpour, Li & Oveis Gharan (JPE 2020)	motivation; waiting not yet tested	H6
Perfect-information benchmark; judge on business outcome	organiser guest session "Building robust ML systems from messy data" (7 Oct 2026)	adopted as diagnostic	§1.4
Grouped hold-out	same session; GroupKFold practice	adopted (seed blocks)	§9.7

Considered and not used:

- tree ensembles and per-user embeddings (too little data; no detected interactions);
- smart defaults (prohibited and uninformative);
- profile-based peer personas (hidden traits are independent of the profile in the generator);
- upper score cutoffs;
- discounting.

9. Baselines, ablations and results

All episodes follow the evaluator's protocol: 60 decision days, then 40 follow-up days. Every policy is run on the **same seeds** (paired).

9.1 Required baselines [dev, seeds 6000-6039 × 6 scenarios, 240 episodes each]

Policy	Realised MSMI /100	Mutual acceptances /100	Coverage
Kit greedy	0.410	5.45	0.335
Kit random-feasible	0.352	5.64	0.337
Kit no-asks	0.117	1.53	0.115

9.2 First ablation [dev, same 240 episodes]

Policy	MSMI /100	Mutual /100 (vs A0)	Coverage (vs A0)
A0 kit greedy	0.410	5.45	0.335
A1 + lexicographic matching	0.408	5.60 (+0.16 ± 0.10)	0.336
A2 + hard-field asker by predicted unlock	0.398	5.74 (+0.29 ± 0.11)	0.338
A3 + scarcity / coverage weights	0.396	5.69 (+0.25 ± 0.10)	0.339
A4 + Thompson learning (old scorer)	0.400	5.64 (+0.19 ± 0.11)	0.340 (+0.005 ± 0.001)

All MSMI differences are within about ± 0.04 (1 SE). The A2 mutual-acceptance gain **did not replicate** on new seeds (§9.9).

9.3 Scorer v2 and tuning round 1 [dev; expected wins per 100 arrived members]

Seed block	Comparison	Expected wins (paired, vs reference)	AUC	Blank clues
7100-7109	kit greedy → v2 + soft asks (≥ 2 options)	0.375 → 0.385 ($+0.0103 \pm 0.0020$, +2.7%)	0.541 → 0.553	70% → 59%
7200-7209	kit greedy → v2 + soft asks (≥ 2 options)	0.360 → 0.383 ($+0.0228 \pm 0.0065$, +6.3%)	0.537 → 0.565	70% → 57%
7200-7209	soft asks for ≥ 1 option (vs ≥ 2)	$+0.0009 \pm 0.0018$	$+0.024 \pm 0.006$	57% → 41%
7200-7209	prior strength 12 (vs 40)	-0.0118 ± 0.0027	-0.009 ± 0.006	-
7200-7209	yes-propensity strength 3 (vs 10)	$+0.0004 \pm 0.0021$	-0.008 ± 0.006	-
7200-7209	yes-propensity off	$+0.0005 \pm 0.0029$	-0.016 ± 0.006	-

Realised mutual acceptances did **not** rise (7100 block: -0.20 ± 0.17 per 100 vs greedy). The expected-wins gain does not come from more first "yes" answers. Block 7300 (\$9.5) suggests it does not come from the reply-behaviour term either; isolating its source is a Round 2 task.

9.4 Allocation: option value and boosts [dev, seeds 7300-7309 × 6 scenarios, 60 paired episodes per policy]

All comparisons are paired against the current best ("B" = scorer v2 + soft asks for ≥ 1 option, with both boosts).

Policy	Expected wins /100	vs B	Coverage vs B	Mutual /100 vs B
Kit greedy	0.3924	-0.0006 ± 0.0047	-0.0018 ± 0.0007	-0.16 ± 0.15
B current best	0.3930	-	-	-
A1 no boosts (plain lexicographic matching)	0.3938	$+0.0007 \pm 0.0021$	-0.0024 ± 0.0006	-0.09 ± 0.08
A2 no coverage boost ($c_{\text{new}} = 1$)	0.3971	$+0.0041 \pm 0.0027$	-0.0013 ± 0.0005	-0.11 ± 0.10
A3 opportunity cost, $\lambda = 0.5$ (no boosts)	0.3980	$+0.0050 \pm 0.0027$	-0.0014 ± 0.0005	-0.08 ± 0.09

- **On this block, B did not beat greedy** ($+0.0006 \pm 0.0047$). This is the third disjoint block; pooled results are in \$0.
- **The coverage boost buys a little coverage (+0.0013) at a likely cost in expected wins (≈ -0.004 , about 1.5 SE).** Optimising a tie-breaker can sacrifice the primary objective. Round 2 default: no coverage boost, or a boost applied only within a narrow score tolerance.
- **Opportunity cost:** after controlling for the removal of the coverage boost, the heuristic showed no clear extra benefit (A3 – A2 paired: $+0.0009 \pm 0.0007$). **H6**

remains open. A one-line penalty is a weak proxy for option value. Round 2 will test explicit one-day look-ahead and selective waiting for contested people.

9.5 Clarification rule and scorer complexity [dev, same 60 episodes]

Variant (vs B, paired)	Expected wins /100	AUC*	Mutual /100	Clarification cost / episode
C1 matching-margin asker	+0.0003 ± 0.0016	−0.024 ± 0.007	−0.02 ± 0.08	225 (B: 274)
S1 simple fixed scorecard (offline prior only; no learning, no reply or yes-propensity terms)	+0.0010 ± 0.0033	+0.027 ± 0.008	+0.34 ± 0.14	273
S2 posterior mean instead of Thompson draws	−0.0029 ± 0.0032	+0.009 ± 0.006	+0.38 ± 0.14	274

*AUC is measured on each policy's own introductions (§5.4), so differences partly reflect which pairs were introduced.

- **Margin asking** reduced clarification spending by about 18% with no detected overall change in expected wins. But it lost in cold start (−0.011 vs broad asking), and lower clarification cost only matters after exact ties on the preceding metrics [rule]. On the fourth block, broad asking beat it (§9.6), so it is **not adopted**.
- **The online learner and Thompson sampling do not earn their place on this evidence.**
 - A fixed scorecard is at least as good on expected wins and better on mutual acceptances (2.4 SE).
 - Posterior means also raise mutual acceptances (2.7 SE). This suggests that exploration costs early introductions it cannot repay within one episode.
 - The reply-behaviour term (H2) showed no measurable benefit in S1.
- These results led to the candidate tested in §9.6.

9.6 Closed-loop feedback and asking rules [dev, new seeds 7400-7419 × 6 scenarios, 120 paired episodes per policy]

R0 = frozen offline scorecard + margin asking + no coverage boost. Errors are **cluster-robust**: 20 seed clusters, because development, delayed and drift share one world per seed.

Policy	Expected wins /100	vs greedy	vs R0	Realised MSMI vs greedy	Ask points
Kit greedy	0.372	–	–	–	198
R0 frozen scorecard, margin asks	0.388	+0.0162 ± 0.0047	–	−0.121 ± 0.048	221
F1 = R0 + population field learning	0.388	+0.0164	+0.0002 ± 0.0017	–	221
F2 = F1 + personal reply / yes learning	0.386	+0.0138	−0.0023 ± 0.0014	–	220
V = R0 with matching-value (VOI) asking	0.389	+0.0175	+0.0013 ± 0.0011†	–	230

Policy	Expected wins /100	vs greedy	vs R0	Realised MSMI vs greedy	Ask points
D0 = R0 without degree boost	0.389	+0.0170	+0.0009 $\pm$ 0.0018†	-	221
R1 = frozen scorecard, broad asks (candidate)	0.392	+0.0205 $\pm$ 0.0051	+0.0043 $\pm$ 0.0019	-0.096 $\pm$ 0.050	270

† plain paired SE.

Does feedback change decisions? On the same state each day, we compared the learner's matching with the frozen scorecard's.

- Population field learning changed the matching on **8.6%** of decision days (4.1% of pairs).
- Adding personal learning raised that to **14.9%** (7.6% of pairs).
- Neither improved expected wins, and personal learning was slightly worse.
- **Conclusion for this simulator: within a 60-day episode, online learning does not repay its variance and delay.** The candidate therefore uses a fixed offline scorecard.

Other findings.

- Broad soft asking beat margin asking (+0.0043 $\pm$ 0.0019).
- A first matching-value (VOI) asker was +0.0013 over margin asking, but was not compared with broad asking. It stays a Round 2 experiment.
- The candidate's gain over greedy is spread across all families except sparse: cold start +0.013, delayed +0.021, development +0.030, drift +0.030, shift +0.027, sparse +0.002.

9.7 Development vs confirmation data

- Seeds 6000–6039, 7000–7029, 7100–7109, 7200–7209, 7300–7309 and 7400–7419 are **development sets**: their results influenced choices. Seeds in the 9000s were used for descriptive analyses.
- **Participant-controlled confirmation block, declared now and untouched:** seeds **50000–50019** for each of the 6 scenario families, 120 episodes. These are public-simulator seeds we control, not equivalent to the organisers' held-back worlds. Once run they are no longer untouched: no tuning on them, and every result will be reported. It will be run once, after the Round 2 policy is frozen, reporting realised MSMI with 95% intervals alongside expected wins, coverage, mutual acceptances, clarification cost, waiting time to first introduction, runtime and invalid-episode count.
- **Pre-declared statistics and power.** On block 7400 the paired standard error over 120 episodes was 0.050 per 100 for realised MSMI and 0.0051 for expected wins. At 80% power these detect differences of about 0.14 ($\approx$ 38% of greedy) and 0.014 ($\approx$ 4%). A 10% realised-MSMI gain would need roughly 1,700 episodes to detect. We therefore declare now: (1) the primary confirmation statistic is the paired expected-wins difference on seeds 50000–50019, which has the same expectation as the realised-MSMI difference (§1.4); (2) realised MSMI on the same 120 episodes is reported with its interval, whatever its sign; (3) if runtime allows,

an extension block, seeds **50020-50299** (1,680 further episodes, also untouched), is run with the same frozen policy so realised MSMI alone can resolve a gain near 10%.

- Seed blocks are **disjoint**, not necessarily independent: within one seed, the development, delayed and drift scenarios share the same generated world, and outcome draws are keyed on (seed, pair, day) [code: `rand_for`]. So per-scenario results within a block are correlated.
- **Multiple comparisons.** We tested many variants, and all are reported. The largest observed development gains are subject to selection bias, so we treat them as provisional until the confirmation run.

9.8 Results by scenario [dev, seeds 7200-7209; expected wins per 100]

Policy	cold start	delayed	development	drift	shift	sparse
Kit greedy	0.357	0.444	0.443	0.417	0.396	0.104
v2 + soft asks (≥ 2)	0.379	0.460	0.482	0.455	0.418	0.103
v2 + soft asks (≥ 1)	0.373	0.457	0.483	0.456	0.431	0.103

- **Sparse shows no gain:** there is almost no choice to exploit.
- On the 7100 block, shift was the weakest scenario for v2 (0.411 vs 0.416 for greedy). There, the development prior points lifestyle the wrong way.
- Clarification cost: about 200 points per episode for greedy, 227 with soft asks (≥ 2), 272 (≥ 1), out of 720.

9.9 Retrospective: what we got wrong

- **An unreplicated gain.** The asker's mutual-acceptance gain ($+0.29 \pm 0.11$) did not repeat on new seeds; an oracle asker gained $\leq \sim 1\%$. *Lesson:* replicate before believing.
- **Discounting "for drift"** was unnecessary under the inspected mechanism (§5.5).
- **Random feasible pairs** matched our full prototype on mutual acceptances. A simple baseline can be as good as a complex model on the outcome that matters, as the guest session stressed.
- **Better prediction $\neq$ more wins.** Broader asking raised AUC by $+0.024 \pm 0.006$ but not expected wins. A clue only helps where it can change a decision.
- **An apparent timing effect.** Introductions made on days 35–49 once appeared to almost never succeed (0 of ~ 430 in two world sets), although their expected success was normal ($\approx 1.0\%$). On 60 fresh worlds the rate was 1.05% (7 of 669): it was chance. Assignment day shows no reliable effect outside the drift scenario's built-in day-35 change.
- **Complexity that did not pay.** The online learner, Thompson sampling, personal habit terms, the coverage boost and the margin asker each looked principled. None beat a simpler alternative (§9.4–9.6).
- **Realised MSMI disagreed with the diagnostic on one block.** On block 7400 greedy scored higher on realised MSMI, while expected wins favoured the candidate by about 4 SE. Rather than explain it away, we tested the diagnostic itself: it matches the simulator exactly, and the pooled gap is within chance (§1.4). The disagreement is outcome luck, and the realised-MSMI interval on 120 episodes (± 0.10 at 95%) is wider than the effect being measured.
- **Corrected early numbers.**

- "20–30 pairs per day" was really 9–19.
- The day-0 allowed-pair count was 2,996, not 3,215 (an age-check bug).
- A "60× faster" claim was $\sim 2\times$ under a fair comparison.

10. Expected failure cases and defences

#	Failure	Defence
F1	Sparse supply (coverage ceiling ≈ 0.16 in sparse)	Prompt matching; global allocation; scarcity and option-value mechanisms under evaluation (H6)
F2	Withheld answers (≈ 65 of 200 decline a hard field)	Never inferred or asked; kept in the denominator
F3	Delayed feedback (replies ≤ 7 days, second answers $\leq \sim 26$ days)	Pending $\neq$ no; prior carries early decisions
F4	Competition for one person	Global matching; scarcity weights
F5	Shifted weights	Fixed scorecard as the robust default. Round 2: a gradual blend of two offline weight sets, weighted by the likelihood of observed replies, kept only if it helps shift without harming the other five families
F6	Noisy personal estimates (~ 2 introductions each)	Strong shrinkage (10 pseudo-observations)
F7	Invalid output	Batch validation (§3); official checker; budget read from state
F8	Timeout (10 s per call, including start-up)	A monotonic-time guard keeps a fixed output margin. If optimisation has not started by the cutoff, a pre-validated simple matching over the already-computed feasible graph is used. If safe completion cannot be guaranteed, an empty batch is returned. Both paths are tested [plan]
F9	Protocol / format faults: NaN or Infinity, NumPy scalar types in JSON, memory near 1 MiB, missing optional keys, empty population, empty feasible graph, all members unavailable, duplicate IDs, members from several pools, declined soft fields	Explicit casting and guards; unit tests for each case [plan]. The policy returns an empty memory object, because the cumulative observable state (introductions, feedback, ask log) is enough to rebuild its statistics on every call; it uses no persistent filesystem state and no information from other episodes
F10	Larger hidden pools	Column filtering; matching within connected components
F11	Over-reading noise	Paired seeds; low-variance diagnostic; untouched confirmation block

11. Observations on the problem framing

1. **Early negative answers do not release people.** Both people stay unavailable for 8 days after an introduction, even after a definitive "no" on day 1 [code]. In a real service, immediate release after a definitive negative answer could increase opportunity without weakening any eligibility constraint.
2. **Thickness more than ranking.** With ~ 3 allowed partners per person and $\sim 75\%$ of them already used by greedy [oracle], improvements are more likely to come

from information (clarification), timing (before departures) and availability than from better pair ranking.

3. **Outcome noise.** With about one qualifying outcome per episode, rankings on realised MSMI over 20 seeds per family will have wide intervals: in our runs, a paired 95% interval of about ± 0.10 per 100 on a baseline of about 0.37, i.e. roughly $\pm 26\%$. We report low-variance diagnostics alongside MSMI for development, while recognising why an official metric should rest on realised, observable outcomes.
4. **Real-service extensions outside the simulator:**
 - retry members who ignored a message;
 - recompute age from date of birth;
 - let users rate the importance of each preference;
 - churn that depends on match experience.

12. Reproducibility and provenance

- **Code and results.** Every number comes from a script in `experiments/`, with raw per-episode JSONL in `experiments/results/`, at commit `d86ff40ecc561157a6c9b15f6b977acc0e556752`.

Section	Script
§9.1–9.2	<code>run_baselines.py</code> , <code>run_ablation.py</code>
§9.3	<code>test_scorer_v2.py</code> , <code>tune_v2.py</code>
§9.4–9.5	<code>tune_v3.py</code>
§9.6	<code>tune_v4.py</code> , <code>tune_v4b.py</code>
§5.2	<code>funnel_probs.py</code> , <code>woe_iv.py</code>
§5.6	<code>scorer_data.py</code>
§1.4	<code>theory_ceilings.py</code> , <code>why_unused.py</code> , <code>daily_cost.py</code> , <code>checks_v3.py</code> , <code>pwin_check.py</code>
Appendix A	<code>bench_100k.py</code>

- **Commands:** run from the organiser kit folder, for example `python tune_v3.py 7300 10 development > out.jsonl`, then `python summarise_scorer_v2.py out*.jsonl`.
- **Environment:** Python 3.13.16, numpy 2.5.3, networkx 3.4.2 (BSD-3). These will be pinned in the Round 2 Docker image.
- **Inference seed:** deterministic per day and phase ($7919 \cdot \text{day} + \text{phase}$).
- **Permitted inputs only.** The policy reads only the observation, clarification results, feedback and its own memory. Hidden values appear only in offline measurement scripts (expected wins, oracle ceilings).
- **AI-assisted tools.** Anthropic Claude was used to help draft and debug experimental scripts, propose and run analyses, prepare mathematical explanations, and edit this research note. The author directed the research questions, reviewed the principal experimental results, and made the final methodological and submission decisions. Source-code interpretations, mathematical arguments and generated prose remain subject to the limitations stated in the note. No external AI service is used by the policy during evaluation.

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- Organiser materials: participant specification 1.0.0 and public kit; session slides "Exploration, Exploitation and Online Learning" (6 Oct 2026) and "Building robust ML systems from messy data" (7 Oct 2026), distributed to participants.

Appendix A. Filter scaling benchmark (outside the evaluated scale)

100,000 synthetic people, 35,051 of them with complete hard fields. All four methods found the identical 7,970,144 feasible pairs [dev, bench_100k.py].

Method	Per new arrival	All pairs
Pair-by-pair row check	21.7 ms	791 s
Group by gender × zone	8.7 ms	289 s
Bitmap index	1.4 ms	35.6 s
Column checks	0.21 ms	6.1 s

In an incremental architecture, newcomer processing is one important scaling path. A production system would also need updates for changed preferences, removals, availability and index repair. At 100k, exact matching would run within connected components or use a greedy approximation.

Appendix B. Scorecard view of the development prior

600 points at even odds; +20 points doubles the odds. One direction starts at 598 points.

Clue	Same	Different	Unknown
goal	+20	−11	0
pace	+10	−10	0
lifestyle	+7	−9	0
conversations	+2	−5	0

Appendix C. Fairness objectives considered

- Utilitarian sum, Nash welfare ($\Sigma \log$), α -fairness, max-min.
- Price of fairness: Bertsimas, Farias & Trichakis (*Operations Research* 2011); Dickerson, Procaccia & Sandholm (AAMAS 2014).
- Group fairness in online matching: Ma, Xu & Xu (AAMAS 2022).
- Code read: FairRec (FairRec_www_2020 @ ad2c455) uses exposure caps plus round-robin turns; FA*IR (fairsearch-fair-python @ a92a3d6) uses minimum quotas.
- Neither uses caps on the best candidates.
- Optional Round 2 tests: $\sqrt{s}$ and $\log s + C$ weights (valid only together with maximum cardinality), and coverage-first weights.

Appendix D. Further material in the repository

- EDGE_CASES.md: 21 edge cases.
- STRESS_TESTS.md: 10 stress-world designs.
- TUNING.md: every tunable parameter, with its reason, range and risk.
- SCORER.md, TABLES.md, NOTES.md: full working notes.